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$$\begin{array}{c|cccc} & 4 & 0 & -7 & -3 \\ 2 & & 8 & 16 & 18 \\ \hline & 4 & 8 & 9 & \boxed{15} \end{array}$$

$\therefore f(2) = 15$ 2

(ii)

$$\begin{array}{c|cccc} & 4 & 0 & -7 & -3 \\ -\frac{1}{2} & & -2 & 1 & 3 \\ \hline & 4 & -2 & -6 & \boxed{0} \end{array}$$ 2

Since $f(-\frac{1}{2}) = 0$, thus $2x+1$ is a factor of $f(x)$.

(iii) $f(x) = (2x+1)(4x^2-2x-6)$

$= 2(2x+1)(2x^2-x-3)$

$= 2(2x+1)(2x-3)(x+1)$ 3

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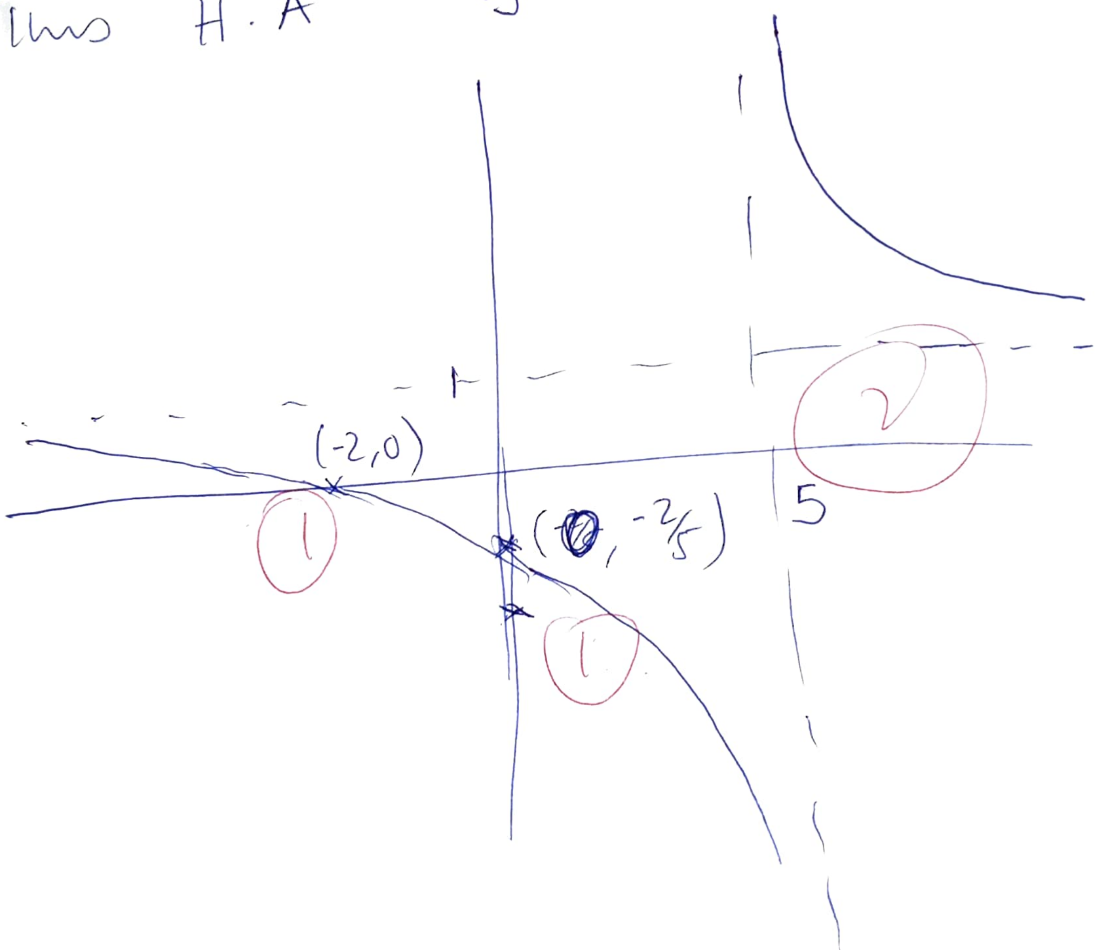
b (i) V. A is $x = 5$ 1

$y = \frac{2+x}{x-5} \Rightarrow xy - 5y = 2 + x \Rightarrow$

$xy - x = 2 + 5y \Rightarrow x(y-1) = 2 + 5y$

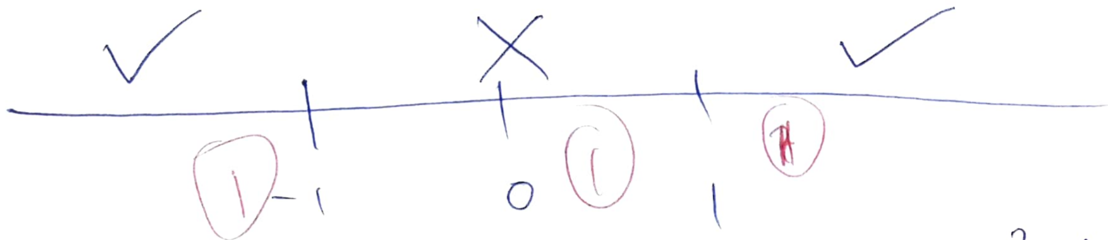
$\Rightarrow x = \frac{2+2y}{y-1}$ 2

Thus H.A $y = 1$



(ii) $\frac{x^2-1}{x^2+1} \geq 0$, Since x^2+1 is always positive, the sign is only affected by numerator

$\therefore x^2 - 1 = (x-1)(x+1)$



Thus testing for $x = -2$, $\frac{x^2-1}{x^2+1} > 0$

When $X = 0$ $\frac{x^2-1}{x^2+1} < 0$

When $x = 2$ $\frac{x^2-1}{x^2+1} > 0$

2

$\therefore$ S.S $= (-\infty, -1] \cup [1, +\infty)$

(ii) $|x+1| = x^2 - 1$ $\Rightarrow$ $x^2 - x - 2 = 0$

① $x + 1 = x^2 - 1$ $\Rightarrow$ $x = -1$

$(x+1)(x-2) = 0$

or $X = 2$

and $x + 1 = -x^2 + 1$ $\Rightarrow$ $x^2 + x = 0$

$x(x+1) = 0$ $\Rightarrow$ $x = 0$ or $x = -1$

$x = 0$ N/A 2

2

only solution for x is $x = -1$ and

$x = 2$ 2

Q 2 b(i) $P = \frac{2\pi}{1/2} = 4\pi$ 2

Amplitude $|-2| = 2$ 2

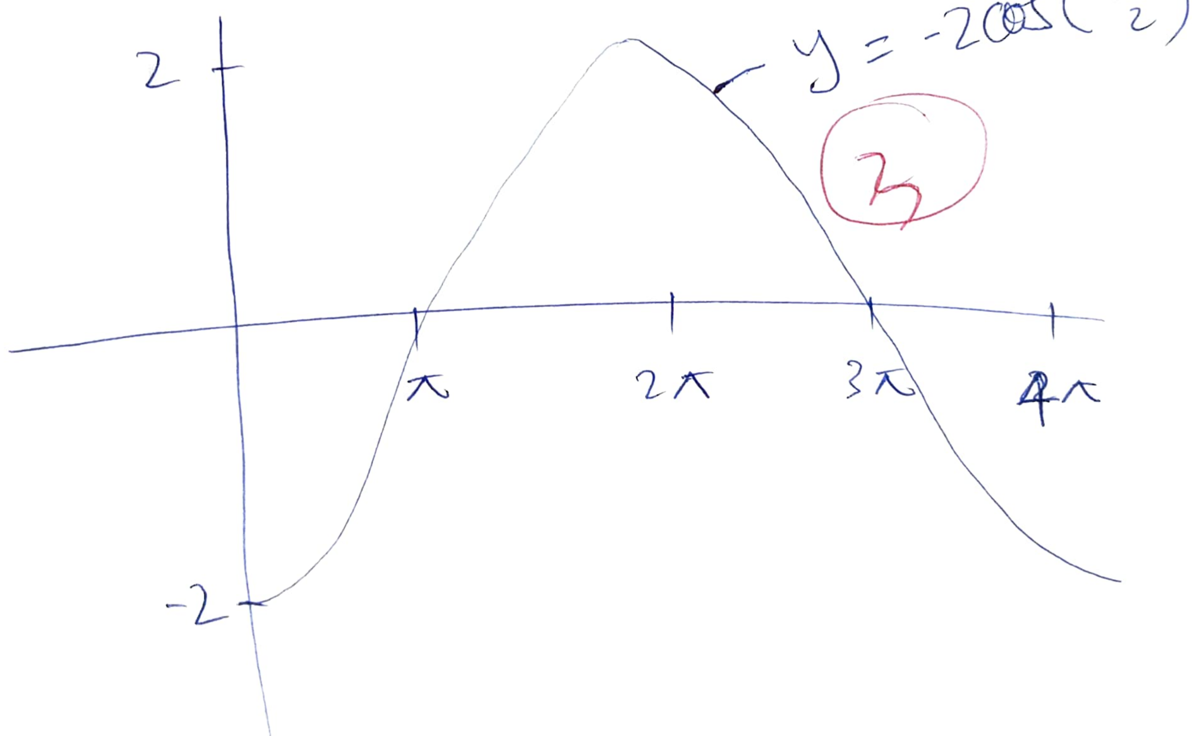
(ii) $0 = -2\cos(-x/2) \Rightarrow \cos(-\frac{x}{2}) = 0$

$\Rightarrow \cos(\frac{x}{2}) = 0 \Rightarrow \frac{x}{2} = \cos^{-1}(0)$

$\frac{x}{2} = 90°, 270°, 450°, 630$

$\Rightarrow x = 180°$ 2 $, 540°$ 2

(iii)



3

(i) $$\frac{\tan x + \tan y}{\cot x + \cot y} = \tan y \tan x$$

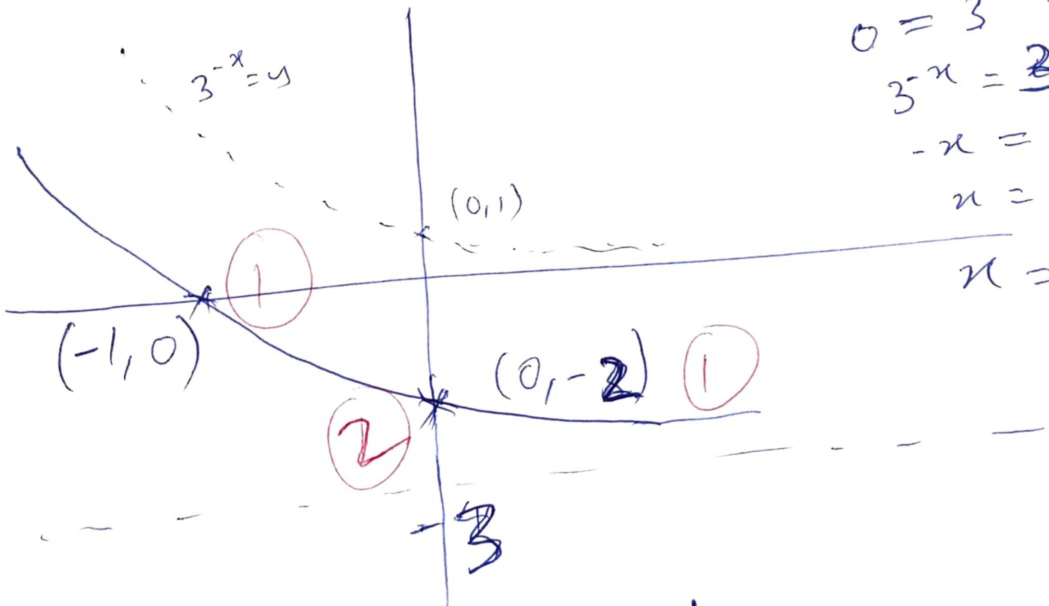
$$L.H.S = \frac{\tan x + \tan y}{\frac{1}{\tan x} + \frac{1}{\tan y}} = \frac{\tan x + \tan y}{\frac{\tan y + \tan x}{\tan x \tan y}}$$

4

$$= \cancel{\tan x + \tan y} \times \frac{\tan x \tan y}{\cancel{\tan y + \tan x}}$$

$$= \tan x \tan y = R.H.S$$

(ii)



$$0 = 3^{-x} - 3$$
$$3^{-x} = 3^1$$
$$-x = \log_3 3$$
$$x = -\log_3 3$$
$$x = -1$$

(iii) $$\frac{x^2}{x^2-1} \Rightarrow x^2-1 \overline{\big) \; x^2} \quad 1$$
$$\underline{-(x^2-1)}$$
$$1$$

$$\therefore \frac{x^2}{x^2-1} = 1 + \frac{1}{x^2-1}$$

2

$\therefore \dfrac{1}{x^2-1} \equiv \dfrac{A}{x-1} + \dfrac{B}{x+1}$

$1 \equiv A(x+1) + B(x-1)$

$\Rightarrow$ if $x = 1$, then $1 = 2A \Rightarrow A = \frac{1}{2}$ (1)

If $x = -1 \Rightarrow -2B = 1 \Rightarrow B = -\frac{1}{2}$ (1)

$\therefore \dfrac{x^2}{x^2-1} = 1 + \dfrac{1}{2(x-1)} - \dfrac{1}{2(x+1)}$

2